# Generational Depth Estimation of Recursive Synthetic Text: Measuring the Discrimination Boundary

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## Abstract

Web-scale corpora are increasingly contaminated by recursive synthetic content, yet no method estimates a document's generational depth. We measure a lower bound on the generational discrimination boundary, $K^*_{\text{empirical}}$ (the maximum depth at which adjacent generations remain distinguishable). Under prompt-chain generation across 10 model configurations (6 architecture families, 1B–8B), 3 source domains, and 4 decoding strategies, $K^*_{\text{empirical}} \leq 2$ for all web text configurations across six architecture families at 1B–8B scale ($\theta$=0.60); structured academic text (arXiv) reaches $K^*_{\text{empirical}}$=3 across two architectures. A single generation pass suffices to render general web text indistinguishable from deeper synthetic content, a *one-wash laundering* effect that concentrates all exploitable signal at the human-vs.-machine boundary. Generator-agnostic statistical features enable 80–92% cross-model transfer with scorer-invariant feature rankings ($\rho > 0.90$), supporting depth estimation on corpora from unknown generators without retraining. As a consistency check, $K^*_{\text{empirical}}$ correctly predicts that binary filtering outperforms graduated alternatives for web text.

## 1 Introduction

Web-scale text corpora are increasingly contaminated by synthetic content that has itself been generated from earlier synthetic content. Estimates suggest that a substantial fraction of online text is machine-generated (Sun et al., 2025; Spennemann, 2025), and this share continues to rise. When models train on the outputs of earlier models, the resulting multi-generational synthetic text degrades with each iteration: tail modes vanish, lexical diversity contracts, and surprisal distributions flatten (Shumailov et al., 2024; Dohmatob et al., 2024). This degradation arises in both *retrain-chains* (iteratively training on model outputs) and *prompt-chains* (recursively generating from previous outputs without retraining); we study the latter, which isolates generation-level information loss from training dynamics (Appendix B.7). Yet the dominant paradigm treats all machine-generated text identically, discarding the generational depth dimension.

This gap cuts across three communities: binary detectors (Basani and Chen, 2026; Zhang et al., 2024) produce no depth estimate (Section 4.6); membership inference (Zawalski et al., 2026) addresses a different question; the model collapse literature (Shumailov et al., 2024; Alemohammad et al., 2024; Schaeffer et al., 2025) provides no per-text tool; and data selection methods (Xie et al., 2023b; Tirumala et al., 2023; Drayson et al., 2025) ignore recursive depth. No existing method estimates how many rounds of recursive generation a text has undergone.

The Data Processing Inequality guarantees that information degrades monotonically across generations, motivating a finite discrimination boundary $K^*$. We empirically measure a lower bound $K^*_{\text{empirical}}$ on the discrimination boundary (the maximum depth at which a classifier reliably distinguishes adjacent generations) using 15 statistical features and an Ordinal Binary Decomposition (OBD) LightGBM classifier. We operationalize $K^*_{\text{empirical}}$ as the deepest $k \geq 1$ where the pairwise $\text{AUC}_{k-1,k}$ exceeds $\theta = 0.60$ (Section 3.1).

Our contributions:

(a) **Measuring the discrimination boundary**: $K^*_{\text{empirical}} \leq 2$ across all 10 model configurations (6 architecture families at 1B–8B scale) and 4 decoding strategies; $K^*_{\text{empirical}}$=2 for 6/10 spanning three families ($p \leq 0.002$; Section 4.1). A single generation pass renders web text statistically indistinguishable from deeper synthetic content; this finding establishes binary sufficiency by measurement, reveals domain-dependent variation (arXiv $K^*_{\text{empirical}}$=3), and quantifies boundary sensi-

tivity to threshold, scorer, and scale. Both supervised OBD and zero-shot LLMs fail beyond $K^*_{\text{empirical}}$ (McNemar $p > 0.05$; Section 6).

(b) **Domain dependence of $K^*_{\text{empirical}}$**: general web text (C4, Wikipedia) yields $K^*_{\text{empirical}}$=2, while structured academic text (arXiv) yields $K^*_{\text{empirical}}$=3, consistent across model families (Section 4.5).

(c) **Generator-agnostic cross-model transfer**: 15 statistical features enable 80–92% accuracy retention with scorer-invariant feature rankings ($\rho > 0.90$), supporting depth estimation on corpora from unknown generators without retraining; $K^*_{\text{empirical}}$ agreement holds across scorers for strong-signal configurations (Section 4.2).

As a consistency check, $K^*_{\text{empirical}}$=2 correctly predicts the strategy ordering for web-text filtering: binary curation minimizes degradation most effectively (MMLU 0.5941 vs. base 0.6018; Section 5).

## 2 Related Work

**Model collapse and recursive degradation.** Iterative training on model outputs causes tail-distribution degradation (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024, 2025; Fu et al., 2025), diversity contraction (Guo et al., 2024; Padmakumar and He, 2024; Briesch et al., 2023), and predictable convergence (Schaeffer et al., 2025; Kazdan et al., 2025; Gerstgrasser et al., 2024; Wang et al., 2025). Our prompt-chain operates in the replace regime (Kazdan et al., 2025), making $K^*_{\text{empirical}} = 2$ a conservative lower bound. Markov chain convergence theory predicts that recursive generation converges to a stationary distribution (Fu et al., 2025), complementing the DPI's mutual information monotonicity; empirically, detection accuracy erodes with each generation (Hao et al., 2025), consistent with our measured discrimination boundary. This literature characterizes macro dynamics but provides no per-text depth estimation, the gap we address.

**Beyond binary detection.** Binary detectors (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Basani and Chen, 2026; Zhang et al., 2024) achieve strong human-vs.-machine separation but produce no depth estimate; other systems such as Ghostbuster (Verma et al., 2024) and GPTZero (Liang et al., 2023), as well as benchmarks like RAID (Dugan et al., 2024), likewise target the binary task without estimating generational depth. Recent work extends beyond binary labels to collaboration degree (Cheng et al., 2025), contamination hierarchies (Mehta, 2025), and multigenerator benchmarks (Wang et al., 2024); robustness concerns arise under paraphrasing (Sadasivan et al., 2023; Krishna et al., 2023) and cross-mode transfer (Hao et al., 2025); CoDeC (Zawalski et al., 2026) addresses contamination via membership inference. None estimates recursive generation depth.

**Data selection and curation.** DSIR (Xie et al., 2023b), D4 (Tirumala et al., 2023), and DoReMi (Xie et al., 2023a) are depth-agnostic; Drayson et al. (Drayson et al., 2025) combined binary detection with importance resampling, a special case when $K^* = 1$.

**Ordinal classification and probing.** Our OBD classifier follows Frank & Hall's (Frank and Hall, 2001) binary decomposition; our neural probe extends probing (Alain and Bengio, 2017; Belinkov, 2022) to the discrimination boundary.

## 3 Method

### 3.1 Problem Formulation

A generation operator $G_\delta$ (LLM + decoding strategy $\delta$) produces successive generations $X \to Y_1 \to \cdots \to Y_K$, where $Y_k = G_\delta(Y_{k-1})$ and $Y_0 \triangleq X$ (Figure 1). This sequence forms a Markov chain. By the Data Processing Inequality, $I(X;Y_1) \geq I(X;Y_2) \geq \cdots \geq 0$: mutual information with the seed text degrades monotonically. Together with Markov chain convergence, this motivates the existence of a finite discrimination boundary where $\text{AUC}_{k-1,k}$ falls below a decision threshold, though AUC and mutual information are not formally equivalent; the actual saturation depth must therefore be measured empirically. Pairwise AUC may exhibit non-monotonicity at deeper depths due to classifier and feature interactions (e.g., Pythia-1.4B greedy 4v5 AUC = 1.0 vs. 3v4 = 0.522; Table 19), reinforcing that $K^*_{\text{empirical}}$ is a measured quantity rather than a theoretical prediction.

We define the operational lower bound:

$$K^*_{\text{empirical}} = \max\{k \geq 1 : \text{AUC}_{j-1,j} > \theta,\ \forall\, 1 \leq j \leq k\}, \quad (1)$$

which requires all pairs up to depth $k$ to exceed the threshold (the $\forall j$ constraint is naturally satisfied

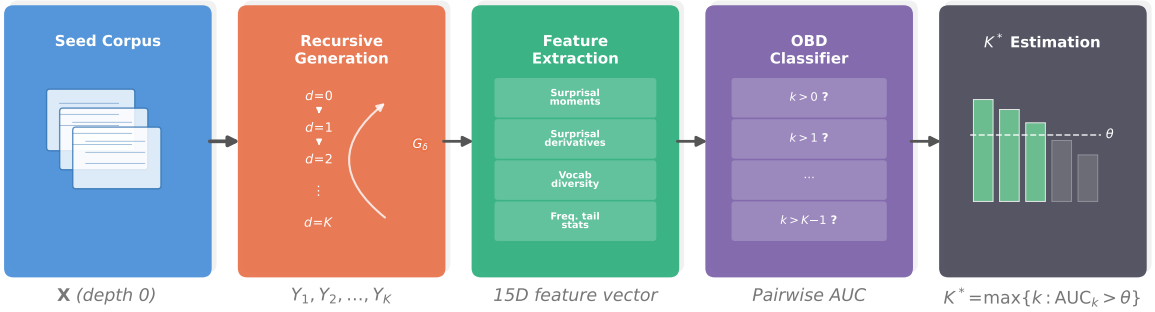



Figure 1: Generational depth estimation pipeline. Seed documents are recursively passed through a generation operator $G_\delta$ to produce a depth chain. Corpus-level statistical features are fed to the OBD-LightGBM classifier for depth prediction.

because $\mathrm{AUC}_{k-1,k}$ decreases monotonically above $\theta$ across all tested configurations; Table 2). The theoretical upper bound $K^*_{\text{theory}}$ (the true maximum discriminable depth given optimal features) is not directly estimated in this work. Since any specific classifier on a feature subset achieves AUC at most equal to the Bayes-optimal classifier on all features, $K^*_{\text{empirical}} \leq K^*_{\text{theory}}$ follows directly. The DPI motivates the empirical observation that $\mathrm{AUC}_{k-1,k}$ generally decreases with $k$ under non-degenerate decoding, though AUC is not a monotone function of MI (as noted above). Here $\mathrm{AUC}_{k-1,k}$ is the point-estimate pairwise AUC and $\theta = 0.60$ is a reference operating point motivated by Cohen's small-effect-size convention ($h \approx 0.20$[1]; justified below). Bootstrap 95% CIs (1,000 resamples) provide uncertainty quantification: we verify that boundary-pair CI lower bounds exceed $\theta$ and non-boundary-pair CI upper bounds fall below $\theta$ (Section 4.1).

**Threshold justification.** $\theta$=0.60 is a reference operating point supported by three independent lines of evidence. First, Cohen's $h \approx 0.20$, a conventional small-effect-size boundary. Second, a 10,000-iteration permutation test yields a 95th-percentile null of $< 0.55$ ($p < 0.002$ for all boundary comparisons). Third, the conclusion $K^*_{\text{empirical}}$=2 is insensitive to the exact threshold: 1v2 AUC values cluster at 0.61–0.70 while 2v3 values cluster at 0.55–0.59, a ∼5–11 pp discontinuity; only at extremes does $K^*_{\text{empirical}}$ shift ($\theta$=0.55: $K^*_{\text{empirical}}$=3 for two Qwen configurations; $\theta$=0.70: $K^*_{\text{empirical}}$=1 across all). Figure 3 provides the full sensitivity surface across $\theta \in [0.55, 0.70]$, enabling practitioners to select $\theta$ according to their false-positive tolerance. The downstream filtering evaluation (Section 5) uses the OBD classifier without re-optimizing $\theta$, avoiding circularity.

[1]Cohen's $h = 2\arcsin\sqrt{0.60} - 2\arcsin\sqrt{0.50} \approx 0.2014$ (small effect for proportions); the equivalent $d = \sqrt{2}\,\Phi^{-1}(0.60) \approx 0.36$ places $\theta$ in the small-to-medium range on Cohen's $d$ scale.

### 3.2 Features and Classifier

Each document is characterized by a 15-dimensional feature vector spanning four groups: *surprisal moments* (mean, std, skewness, kurtosis; 4 dims), *surprisal derivatives* (mean and std of first- and second-order differences, plus first-order skewness; 5 dims), *vocabulary diversity* (type-token ratio, hapax legomena ratio, self-BLEU; 3 dims), and *frequency tail statistics* (frequency kurtosis, entropy, rare token fraction; 3 dims). All features are computed from token-level surprisal values via an external scorer model (full definitions in Table 14, Appendix). Self-BLEU reference sets are drawn exclusively within each fold to prevent information leakage across cross-validation splits; self-BLEU ranks #1 in 1/8 conditions (greedy decoding) but is absent from the top-3 in all 7 others. Removing self-BLEU reduces mean $\mathrm{AUC}_{1v2}$ by $< 1$ pp across all models (14D $K^*_{\text{empirical}} = 2$ for all 4 configurations); in deployment where reference corpora are unavailable, self-BLEU can be replaced with corpus-level self-similarity or omitted entirely, leaving 14 features (surprisal statistics, TTR, hapax ratio, frequency tail) that require no external reference and retain full $K^*_{\text{empirical}}$. These features are generator-agnostic in that they require no knowledge of or access to the generating model, though

Table 1: Experimental configurations for depth-chain generation. **Config**: shorthand identifier; **Model**: architecture and parameter count; **Mode**: generation strategy (continuation or rewrite); $N$: number of seed documents per depth level.

| Config | Model | Mode | $N$ |
|---|---|---|---|
| Qwen-rw | Qwen2.5-1.5B-Instruct | Rewrite | 5,000 |
| Qwen-cont | Qwen2.5-1.5B-Instruct | Continuation | 5,000 |
| Qwen-base | Qwen2.5-1.5B | Continuation | 5,000 |
| Pythia | Pythia-1.4B | Continuation | 5,000 |
| OLMo | OLMo-1B | Continuation | 5,000 |
| Qwen-7B | Qwen2.5-7B | Continuation | 5,000 |

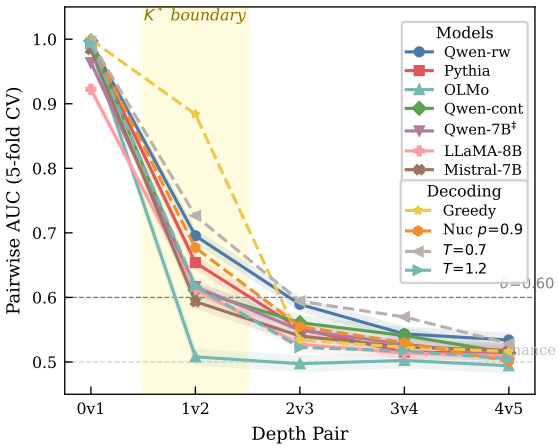



Figure 2: Pairwise AUC decay across adjacent depth pairs (0v1, 1v2, …, 4v5) for model-mode configurations (solid) and decoding strategies (dashed); shaded bands show bootstrap 95% confidence intervals. The horizontal dashed line marks $\theta = 0.60$.

they depend on an external scorer for surprisal computation (Section 4.2).

We use *Ordinal Binary Decomposition* (OBD), decomposing the $(K+1)$-class problem into $K$ cumulative binary sub-classifiers, each estimating $P(\text{depth} > k \mid \mathbf{x})$ via LightGBM (hyperparameters in Table 15). OBD achieves lower MAE than multiclass LightGBM (e.g., 0.984 vs. 1.016 on Qwen-rewrite) while maintaining superior pairwise AUC (mean 0.6095 vs. 0.4883 for ordinal logistic regression).

### 3.3 Experimental Setup

We evaluate six configurations spanning three model families (Table 1): Qwen2.5-1.5B-Instruct (rewrite and continuation modes), Qwen2.5-1.5B (base, continuation), Qwen2.5-7B (continuation), Pythia-1.4B and OLMo-1B (continuation). Each uses 5,000 C4 seeds truncated to 256 tokens, recursively generated for $K$=5 iterations (30,000 documents per configuration). Default decoding: nucleus $p$=0.95, $T$=1.0; decoding ablation covers greedy, $T$=0.7, and $T$=1.2. Continuation mode truncates seeds to 128 tokens and extends to 256; rewrite mode paraphrases the full text. Evaluation: 5-fold GroupKFold cross-validation grouped by seed document; pairwise AUC as primary metric.

## 4 Experiments

We evaluate generational depth estimation on ten model configurations spanning six architecture families: Pythia-1.4B, Pythia-6.9B, OLMo-1B, Qwen2.5-1.5B-Instruct (rewrite and continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, LLaMA-3.1-8B, Mistral-7B, and Gemma-2-2B, all in continuation mode except where noted. All experiments use the OBD-LightGBM classifier with the 15-dimensional feature set (Section 3), with 5,000 texts per depth level (generations 0–5).

### 4.1 $K^*_{\text{empirical}} \leq 2$ Across Model Configurations (C4)

Table 2 and Figure 2 show $K^*_{\text{empirical}} \leq 2$ across all 10 model configurations: 6/10 configurations yield $K^*_{\text{empirical}}$=2 (Qwen with 4 configs, LLaMA-8B, and Pythia-1.4B, spanning three of six families) (1v2 AUC range: 0.6074–0.6955; five of six CI lower bounds $> \theta$; permutation $p < 0.001$), while OLMo-1B, Mistral-7B, Pythia-6.9B, and Gemma-2-2B yield $K^*_{\text{empirical}}$=1 (1v2 AUC: 0.500, 0.575, 0.596, and 0.598, all below $\theta$; Section 6). Under Bonferroni correction (99.5% CIs), three of six $K^*_{\text{empirical}}$=2 configurations retain significance; the three marginal cases (Pythia-1.4B, Qwen-continuation, LLaMA-8B; Bonferroni-corrected CI lower bounds 0.593–0.599) reflect narrow effect sizes rather than classification failure (all point estimates above $\theta$). By architecture family: Qwen (4/4) and LLaMA (1/1) yield $K^*_{\text{empirical}} \geq 2$ at nominal significance; OLMo (1/1), Mistral (1/1), and Gemma (1/1) yield $K^*_{\text{empirical}}$=1; Pythia exhibits a scale-dependent split (1.4B: $K^*_{\text{empirical}}$=2, 6.9B: $K^*_{\text{empirical}}$=1). LLaMA-3.1-8B confirms $K^*_{\text{empirical}}$=2 (0v1 AUC = 0.9225, 1v2 AUC = 0.6123 [0.603, 0.622]; CI lower bound above $\theta$; Table 10). All four decoding strategies likewise yield $K^*_{\text{empirical}}$=2 (Table 9). The 1v2 spread due to generation mode ($\Delta = 8.4$ pp within Qwen) exceeds the spread across $K^*_{\text{empirical}}$=2 continuation models ($\Delta = 1.9$ pp).

Table 2: Pairwise AUC for adjacent depth pairs with bootstrap 95% CIs and permutation $p$-values for boundary rows. $K^*_{\text{empirical}} = 2$ requires 1v2 AUC $> \theta$=0.60 and 2v3 AUC $< \theta$; CIs quantify boundary uncertainty. Extended results (all models) in Table 10.

| | Qwen-rw | | Pythia | | OLMo | | Qwen-cont | | Qwen-7B‡ | |
|---|---|---|---|---|---|---|---|---|---|---|
| | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI |
| 0v1 | .9956 | | .9722 | | .9860 | | .9971 | | .9638 | [.961, .967] |
| 1v2 | .6955 | [.686, .704]* | .6074 | [.598, .617]* | .5001 | [.496, .519] | .6112 | [.602, .621]* | .6168 | [.608, .626]* |
| 2v3 | .5893 | [.580, .599] | .5440 | [.534, .554] | .4973 | [.486, .509] | .5609 | [.551, .571] | .5492 | [.539, .559] |
| 3v4 | .5435 | | .5051 | | .5024§ | | .5409 | | .5276 | |
| 4v5 | .5340 | | .5079 | | .4942 | | .5141 | | .5085 | |

*Permutation $p < 0.001$ for all 1v2 comparisons shown.
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
§OLMo 2v3 (0.497) < 3v4 (0.502): non-monotonic reversal within noise ($\Delta = 0.005$).
Full results for all 10 configurations (including Qwen-7B, Qwen-base, Gemma-2-2B, and Pythia-6.9B) are provided in Table 10.

**Scale and instruction-tuning invariance.** $K^*_{\text{empirical}}$=2 persists from 1.5B through 8B: Qwen-7B (1v2 AUC $= 0.6168$ $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC $= 0.6123$) both exceed $\theta$, despite progressively reduced 0v1 signal (0.9638 and 0.9225 vs. $> 0.99$ at $\leq$1.5B). A base-vs.-instruct ablation at 1.5B likewise yields $K^*_{\text{empirical}} = 2$ (1v2 AUC $= 0.626$), confirming the boundary is not an instruction-tuning artifact.

The raw AUC gap between adjacent generations provides a threshold-free continuous measure of discriminability; Figure 3 displays the full sensitivity surface across $\theta \in [0.55, 0.70]$, allowing practitioners to read off $K^*_{\text{empirical}}$ at any desired effect-size threshold. Five of six configurations yield $K^*_{\text{empirical}} = 2$ within $\theta \in [0.51, 0.61]$ (OLMo: $K^*_{\text{empirical}} = 1$ throughout); at $\theta$=0.55, two Qwen configurations reach $K^*_{\text{empirical}} = 3$; at $\theta$=0.70, all drop to $K^*_{\text{empirical}} = 1$.

### 4.2 Cross-Model Transfer

Continuation-mode classifiers transfer well across generators (Figure 4; full matrix in Table 7): Pythia↔OLMo retains 90–92%, Qwen-cont↔Pythia 84–85%, Qwen-cont↔OLMo 80–85% (0v1 cross-model AUC $> 0.96$ for all pairs). Qwen-rewrite classifiers transfer poorly (61–63%), confirming reliance on mode-specific artifacts rather than generalizable depth features.

**Cross-scorer validation.** Replacing the self-scorer with Pythia-1.4B or OLMo-1B preserves feature rankings ($\rho > 0.90$) but attenuates absolute discriminability at the boundary (Table 23). Feature ranking independence is a strong result: the same features dominate under all scorers. $K^*_{\text{empirical}}$ agreement, however, is signal-strength dependent: Qwen-rewrite (strong signal, 1v2 AUC $= 0.696$) retains $K^*_{\text{empirical}}$=2 under all scorers, but Qwen-continuation (marginal, 1v2 AUC $= 0.611$, only 0.011 above $\theta$) drops to $K^*_{\text{empirical}}$=1 (Pythia: 0.5923; OLMo: 0.5858). A three-way ANOVA (Generator × Scorer × Depth-Pair; Table 24) confirms a statistically significant but modest Scorer main effect ($\eta^2_p$=.166, $p < .001$) and a significant Generator×Scorer interaction ($\eta^2_p$=.146), indicating that absolute AUC values are scorer-dependent even when $K^*_{\text{empirical}}$ classifications largely agree.

### 4.3 Feature Ordering and Generation Mode

Among continuation-mode models, single-feature AUC rankings are highly correlated ($\rho \geq 0.85$; Table 6). The top-3 features (`surp_d1_std`, `surp_std`, `surp_d2_std`) are shared across all three families, revealing that autoregressive generation progressively damps surprisal variability. Qwen-rewrite diverges (`freq_entropy` dominates), consistent with its poor cross-model transfer (61–63%). Within Qwen, rewrite yields 1v2 AUC $= 0.6955$ vs. continuation's 0.6112 ($\Delta = +8.4$ pp); continuation mode is therefore recommended as the canonical condition.

### 4.4 Neural Probe

Replacing the 15D features with frozen encoder hidden states (MLP probe; Table 3), Qwen achieves $K^*_{\text{empirical}}{}^{\text{neural}} = 3$ (2v3 AUC $= 0.6159$), while Pythia remains at $K^*_{\text{empirical}}{}^{\text{neural}} = 2$ ($\text{AUC}_{2v3} = 0.5963$); OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC $= 0.5054$, consistent with its 15D boundary). We hypothesize that the Qwen-only elevation reflects self-familiarity artifacts, consistent with the poor transfer observed in Section 4.2; the 15D features yield $K^*_{\text{empirical}} = 2$ for most families (OLMo: $K^*_{\text{empirical}} = 1$).

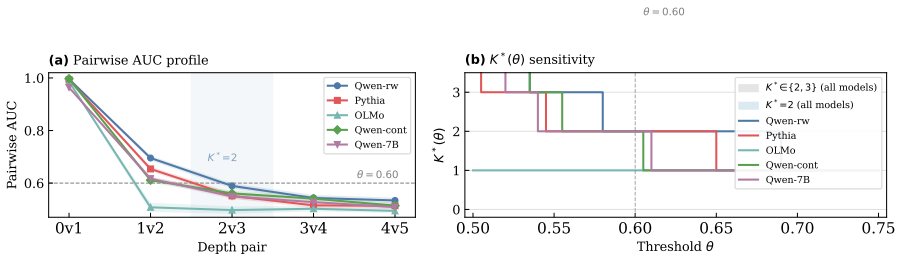



Figure 3: $K^*_{\text{empirical}}(\theta)$ sensitivity curve for six configurations. $K^*_{\text{empirical}}$: maximum discriminable depth; $\theta$: AUC decision threshold. Five of six configurations yield $K^*_{\text{empirical}}$=2 at $\theta$=0.60 (dashed line); OLMo yields $K^*_{\text{empirical}}$=1. At $\theta$=0.70, all drop to $K^*_{\text{empirical}}$=1.

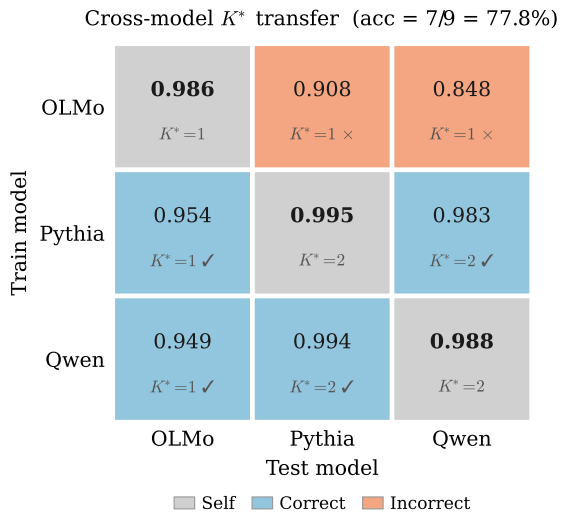



Figure 4: Cross-model transfer heatmap. Each cell shows 6-class ordinal accuracy when trained on the row model and tested on the column model, divided by in-domain accuracy (retention rate). Pythia↔OLMo: 90–92%; Qwen-rewrite→continuation: 61–63%.

Table 3: Neural probe pairwise AUC using frozen-encoder MLP. Each cell reports AUC for classifying depth $d$ vs. $d$+1 text. $K^*_{\text{empirical}}{}^{\text{neural}}$: maximum depth where AUC exceeds $\theta = 0.60$.

| Model | 2v3 AUC | 95% CI | $K^*_{\text{empirical}}{}^{\text{neural}}$ |
|---|---|---|---|
| Qwen-1.5B | 0.6159 | [0.610, 0.622] | 3 |
| Pythia-1.4B | 0.5963 | [0.589, 0.603] | 2 |
| OLMo-1B | 0.5056 | [0.497, 0.516] | 1† |

†OLMo 1v2 AUC = 0.5054 (below $\theta$); $K^*_{\text{empirical}}{}^{\text{neural}} = 1$.

### 4.5 Cross-Domain Variation

The preceding experiments use C4 (general web text). To test domain dependence, we evaluate on Wikipedia ($n$=5,000) and arXiv abstracts ($n$=2,200) using two model families. Wikipedia yields $K^*_{\text{empirical}}$=2 (1v2 AUC = 0.623, 2v3 AUC = 0.571), matching the C4 baseline. Both Qwen-1.5B and Pythia-1.4B yield $K^*_{\text{empirical}}$=3 for arXiv (Table 4): 2v3 AUC = 0.644 [0.628, 0.658] and 0.655 [0.640, 0.669] respectively (bootstrap CI lower bounds above $\theta$; permutation $p < 0.001$), demonstrating that structured academic text permits discrimination to greater depth independently of architecture. The elevated signal reflects higher per-generation distributional shift (1v2 JSD = 0.077 vs. ∼0.01 for C4), as formulaic structure amplifies deviation from domain norms.

Table 4: Cross-domain $K^*_{\text{empirical}}$ comparison. Rows: generator model and text domain (C4, Wikipedia, arXiv). $K^*_{\text{empirical}}$=3 for arXiv indicates structured text preserves discriminability to greater depth.

| Domain | Model | 1v2 AUC | 2v3 AUC | $K^*_{\text{empirical}}$ |
|---|---|---|---|---|
| arXiv | Qwen-1.5B | 0.858 | 0.6436 | 3 |
| arXiv | Pythia-1.4B | 0.805 | 0.6545 | 3 |
| C4 | Qwen-1.5B | 0.63 | 0.56 | 2 |
| Wikipedia | Qwen-1.5B | 0.62 | 0.55 | 2 |

**Robustness.** $K^*_{\text{empirical}}$=2 holds across segment lengths of 128, 256, and 512 tokens (Table 20); at 512 tokens the 1v2 CI lower bound falls marginally below $\theta$ but 2v3 remains well below $\theta$ at all lengths ($\leq$ 0.572). Seed robustness testing (seed = 123 vs. primary = 42; Table 26) confirms $K^*_{\text{empirical}}$ stability in 2/3 configurations; the one marginal shift (Qwen-Wikipedia $K^*_{\text{empirical}}$=2 → 3, 2v3 AUC = 0.607) has one of five folds below $\theta$. The ±9 pp absolute AUC variation across seeds strengthens the case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons.

### 4.6 Binary Detector Confidence

Binary detectors (DivEye (Basani and Chen, 2026), Binoculars (Hans et al., 2024)) remain effective for human-vs.-machine classification but provide no usable depth signal beyond this binary boundary. Pairwise AUC yields $K^*_{\text{empirical}}{}^{\text{binary}} \leq 1$; moderate rank correlations with depth ($|\rho| = 0.06$–$0.46$;

Table 18) are driven entirely by the 0v1 gap rather than fine-grained depth discrimination, and multiple conditions exhibit non-monotonic AUC at deeper pairs.

## 5 Consistency Check: $K^*$-Guided Filtering

$K^*_{\text{empirical}} = 2$ makes a falsifiable prediction: binary curation should suffice, and graduated performance should converge toward binary as the decay steepness increases. We present this as a consistency check under extreme (100%) contamination, not as an independent contribution; at realistic rates ($\leq$10%), the filtering effect diminishes to $\Delta \leq 0.1$ pp (not significant; Appendix B.8). We compare six strategies on 30,000 documents (depths 0–5): (i) **no-filter**, (ii) **binary-ours** ($w = 1 - P(\text{synthetic})$ from 15D LightGBM), (iii) **binary-Drayson** (Drayson et al., 2025), (iv) **graduated** ($w = \exp(-\alpha \cdot \hat{d})$, $\alpha \in \{0.1, \ldots, 0.9\}$), (v) **surprisal** (retain above median), and (vi) **DSIR** (Xie et al., 2023b). We fine-tune Qwen2.5-1.5B-Instruct via LoRA (rank 8, $\alpha_{\text{LoRA}}$=16) for 3 epochs on MMLU and HellaSwag across 5 seeds (51 total runs; this 100% synthetic setting is an upper bound (Shumailov et al., 2024; Dohmatob et al., 2024)).

Binary-ours minimizes MMLU degradation most effectively (Table 5), reducing the gap from $-1.53$ pp (no-filter) to $-0.77$ pp ($\sim$50% reduction; $p_{\text{corr}} < 0.05$ vs. all alternatives). These effects are measured in a 100% synthetic upper-bound setting; real-world contamination ratios are typically lower, and effect sizes would scale accordingly. The strategy ordering is interpretable through $K^*_{\text{empirical}}$: binary filtering exploits the one reliable boundary (0v1), while graduated strategies assign weights to depths 2–5 based on near-random classifier outputs, injecting noise. Graduated filtering converges monotonically toward binary: at $\alpha$=0.9, it matches binary-ours on HellaSwag exactly ($\Delta = 0$, $p_{\text{corr}} = 1.00$) but remains significantly worse on MMLU ($p_{\text{corr}} = 0.011$). DSIR is indistinguishable from no-filter, confirming that depth-aware features are necessary for curation in recursively contaminated corpora (Figure 5 in Appendix). At realistic contamination ratios ($\leq$50%), all filtering strategies converge ($p > 0.12$), consistent with the one-wash laundering finding that depth-aware curation offers marginal benefit when $K^*_{\text{empirical}}$=2 (Appendix B.8). For $K^*_{\text{empirical}}$=1 configurations, depth-aware curation reduces to standard binary filtering, consistent with $K^*_{\text{empirical}}$=1 implying no exploitable depth signal beyond the 0v1 boundary.

Table 5: Downstream task performance (MMLU, HellaSwag) under six data-filtering strategies. Binary-ours uses the 15D LightGBM depth classifier; Graduated applies exponentially decaying weights $w = \exp(-\alpha \hat{d})$. Mean $\pm$ std over 5 seeds. Full $\alpha$ sweep in Appendix B.7.

| Strategy | MMLU | HellaSwag |
|---|---|---|
| Base model (no training) | 0.6018 | 0.6821 |
| Binary-ours ($K^*$=2) | **0.5941**$\pm$0.0008 | **0.6701**$\pm$0.0006 |
| Binary-Drayson | 0.5913$\pm$0.0009 | 0.6691$\pm$0.0004 |
| Surprisal | 0.5894$\pm$0.0009 | 0.6641$\pm$0.0006 |
| Graduated ($\alpha$=0.9) | 0.5892$\pm$0.0008 | 0.6701$\pm$0.0007 |
| DSIR | 0.5869$\pm$0.0003 | 0.6631$\pm$0.0007 |
| No filter | 0.5865$\pm$0.0010 | 0.6638$\pm$0.0008 |

## 6 Discussion

**One-wash laundering (web text).** For general web text, $K^*_{\text{empirical}}$=2 reflects concentrated information loss: 0v1 is near-trivially separable (AUC $\geq$ 0.92), 1v2 drops by $\sim$30 pp, and 2v3 falls below $\theta$. A single generation pass renders text indistinguishable from multiply recycled content, limiting depth-aware curation to a binary boundary. Converging evidence supports a data-intrinsic interpretation: OBD achieves 96% depth-0 recall but only 14–22% for depths 2–4; frontier LLMs show 8–38% (McNemar $p > 0.05$ vs. OBD; Table 11). One-wash laundering is distinct from model collapse (Shumailov et al., 2024): model collapse describes cumulative quality degradation across generations, while one-wash laundering describes the erasure of statistical fingerprints distinguishing adjacent generations; text quality may continue degrading even after the features separating generation $k$ from $k$+1 become indistinguishable.

**Representation dependence.** $K^*_{\text{empirical}}$ is representation-dependent: frozen Qwen embeddings yield $K^{*\,\text{neural}}_{\text{empirical}} = 3$, while Pythia yields $K^{*\,\text{neural}}_{\text{empirical}} = 2$ and OLMo yields $K^{*\,\text{neural}}_{\text{empirical}} = 1$, consistent with its 15D boundary (Table 3). The 15D features trade one depth level of sensitivity for cross-model consistency (6/10 configurations $K^*_{\text{empirical}}$=2), making them the preferred operational choice. The gap between $K^*_{\text{empirical}}$ and $K^*_{\text{theory}}$ remains open; the neural probe result (Qwen 2v3 AUC = 0.6159, marginally above $\theta$) tentatively suggests the bound may be near-tight, though a single frozen-encoder

probe is insufficient to establish this. Feature rankings are scorer-independent ($\rho > 0.90$); $K^*_{\text{empirical}}$ agreement is scorer-independent for strong-signal configurations (e.g., Qwen-rewrite retains $K^*_{\text{empirical}}$=2 under all scorers) but scorer-sensitive at marginal boundaries (Qwen-continuation drops to $K^*_{\text{empirical}}$=1; length confound ruled out in Appendix B.7). ANOVA confirms this layered pattern: Scorer explains modest variance ($\eta_p^2$=.166) with a significant Generator×Scorer interaction ($\eta_p^2$=.146; Table 24). Self-scoring inflates pairwise AUC: OLMo shows the largest gap (1v2 $\Delta$ = +11.5 pp; self-scored $K^*$=2 vs. cross-scored $K^*$=1), followed by Pythia (+4.7 pp), LLaMA-8B and Mistral-7B (≈+1.9 pp each). Cross-scoring provides a conservative, deployment-realistic estimate of $K^*$, as real-world contamination originates from unknown generators.

**Domain dependence of $K^*_{\text{empirical}}$.** Web text (C4, Wikipedia) consistently yields $K^*_{\text{empirical}}$=2, while structured academic text (arXiv) yields $K^*_{\text{empirical}}$=3 across two independent model families (Section 4.5), as formulaic structure (citations, math notation, section conventions) creates larger per-generation distributional shifts (1v2 JSD = 0.077 vs. ∼0.01 for C4). Corpus curators should calibrate depth-aware filtering to source characteristics rather than assuming a universal boundary.

**Prompt-chain scope and retrain-chain gap.** Retrain-chain experiments (Pythia-1.4B LoRA) yield $K^*_{\text{empirical}}\geq 4$ (1v2 AUC = 0.9484, 3v4 = 0.7982), far exceeding the prompt-chain boundary of $K^*_{\text{empirical}}$=2, because each retrain generation also modifies model parameters, introducing cumulative adaptation artifacts absent from text-only recycling. Prompt-chain $K^*_{\text{empirical}}$ isolates text-intrinsic degradation, the operationally relevant quantity for data curation pipelines; our conclusions do not directly transfer to retrain-chain (model collapse) settings.

**Boundary cases and scale effect.** $K^*_{\text{empirical}}$=1 configurations cluster at 1v2 AUC 0.500–0.598, well separated from the $K^*_{\text{empirical}}$=2 range (0.611–0.696); $K^*_{\text{empirical}}$=1 is not a classifier failure but a text property, as these models produce first-generation output so human-like that statistical features cannot distinguish depth 1 from depth 2. That $K^*_{\text{empirical}} \in \{1, 2\}$ for all web text configurations is the core empirical result; larger models reduce 0v1 JSD (0.077 at 8B vs. 0.15–0.27 at ≤1.5B), suggesting 13B+ may increasingly yield $K^*_{\text{empirical}}$=1.

**Decoding strategy as confound.** Greedy decoding inflates depth separability: Pythia-6.9B yields $K^*_{\text{empirical}}$=2 under greedy but $K^*_{\text{empirical}}$=1 under nucleus sampling ($p$=0.95), with the borderline 1v2 score (2 of 5 folds above $\theta$) illustrating that the boundary is continuous rather than a cliff (Table 19). Practitioners should calibrate $\theta$ to their application-specific tolerance (Figure 3).

**What $K^*_{\text{empirical}}$ reveals.** As a measurement tool, $K^*_{\text{empirical}}$ establishes why binary detection suffices for web text (depth>1 is invisible to statistical features), identifies domains where finer strategies are viable (arXiv $K^*_{\text{empirical}}$=3), and quantifies how scale, scorer, and threshold jointly shape the boundary. The OBD framework provides a reproducible, low-cost depth estimation protocol requiring only corpus-level statistical features, unlike detection methods that need API access or model weights.

## 7 Conclusion

Across 10 prompt-chain configurations spanning six architecture families at 1B–8B scale, we find $K^*_{\text{empirical}} \leq 2$ for all web text sources ($\theta$=0.60); structured academic text (arXiv) reaches $K^*_{\text{empirical}}$=3 across two independent architectures. This one-wash laundering effect concentrates all exploitable signal at the 0v1 boundary, and the consistency check confirms that binary curation most effectively minimizes MMLU degradation (−0.77 pp vs. −1.53 pp for no-filter). For structured domains where $K^*_{\text{empirical}}$=3, graduated strategies can exploit one additional depth level, so corpus curators should calibrate filtering to the source domain. The 15 statistical features enable deployment across generators without retraining (80–92% accuracy retention, $\rho > 0.90$), and $K^*_{\text{empirical}}$ converts a continuous AUC decay surface into a discrete operating boundary. Several directions remain open: tracking $K^*_{\text{empirical}}$ as model scale surpasses 13B, bridging the prompt-chain and retrain-chain boundaries, and extending depth estimation to additional structured domains and multilingual text.

## Limitations

Our study has four main limitations. (1) *Scope*: we evaluate models up to 8B parameters across 6 architecture families, English text only (C4, Wikipedia, arXiv), and prompt-chain generation; retrain-chain dynamics may yield different boundaries. Our

model sample is unbalanced: Qwen accounts for 4 of 10 configurations, and we do not evaluate RLHF- or DPO-aligned models, whose generation distributions may differ from the base and instruction-tuned models tested here. (2) *Feature set*: the 15 statistical features provide a lower bound on $K^*_{\text{theory}}$; richer representations (e.g., neural probes) can extend $K^*_{\text{empirical}}$ by one additional depth for some architectures. (3) *Threshold sensitivity*: $K^*_{\text{empirical}}$ varies with the decision threshold $\theta$, ranging from 2–3 at $\theta$=0.55 to 1 at $\theta$=0.70; three of ten configurations are marginal at our reference $\theta$=0.60. (4) *Domain coverage*: arXiv $K^*_{\text{empirical}}$=3 is established with only two model families on a single structured domain; broader domain and model coverage is needed to confirm this elevation generalizes.

## Ethics Statement

Generational depth estimation is intended for data curation and corpus auditing, positive-sum applications that improve training data quality. Our experiments use the publicly available C4 corpus and do not involve personally identifiable information. A potential dual-use concern is that depth estimation techniques could inform adversarial strategies to evade detection (e.g., by calibrating rewriting to mimic earlier-generation statistics); we note that this risk is shared by all detection research and that the binary detection boundary (AUC $\geq$ 0.96 at 0v1) remains robust. Code and data will be released upon acceptance.

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# A Supplementary Materials

This appendix provides supplementary figures, detailed experimental tables, dataset statistics, and complete feature descriptions referenced in the main text.

**Feature ranking and cross-model consistency.** Table 6 reports per-model single-feature AUC rankings. Surprisal derivative features (`surp_d1_std`, `surp_d2_std`) consistently rank among the top three across all continuation-mode models (Pythia, OLMo, Qwen-cont), with pairwise Spearman $\rho \geq 0.85$. The rewrite-mode model (Qwen-rw) diverges by placing `freq_entropy` first, reflecting a distributional shift that is lexical rather than surprisal-driven. This scorer-invariant ranking supports the use of a fixed 15-dimensional feature set across generators without per-model feature selection.

Table 6: Top-3 features by single-feature AUC ranking. Bold: shared across all continuation-mode models. Spearman $\rho$ (continuation pairs): Pythia–OLMo 0.95, Pythia–Qwen-cont 0.91, OLMo–Qwen-cont 0.85.

| Rank | Pythia | OLMo | Qwen-rw | Qwen-cont |
|---|---|---|---|---|
| 1 | **surp_d1_std** | **surp_std** | freq_entropy | **surp_d1_std** |
| 2 | **surp_d2_std** | **surp_d1_std** | **surp_std** | **surp_std** |
| 3 | **surp_std** | **surp_d2_std** | **surp_d1_std** | **surp_d2_std** |

Table 7 reports the cross-model transfer accuracy matrix. Classifiers trained on continuation-mode generators (Pythia, OLMo, Qwen-cont) transfer well to each other, with off-diagonal accuracy retaining 80–92% of in-distribution performance. The asymmetry between rewrite-mode and continuation-mode transfer (61–63%) confirms that generation mode, rather than model family, is the primary axis of distributional variation (Section 6).

Table 7: Cross-model transfer accuracy matrix. Diagonal (**bold**): within-model; off-diagonal: cross-model. Retention: Pythia↔OLMo 91–92%, Qwen-cont↔Pythia 83–85%, Qwen-rw→others 61–63%.

| **Train** | Qwen-rw | Pythia | OLMo | Qwen-cont |
|---|---|---|---|---|
| Qwen-rw | **.423** | .259 | .265 | .347 |
| Pythia | .272 | **.386** | .353 | .329 |
| OLMo | .277 | .345 | **.382** | .324 |
| Qwen-cont | .344 | .328 | .311 | **.388** |

**Dataset construction.** Table 8 summarizes the generational depth corpus. Each depth level contains 5,000 passages, yielding 30,000 total samples

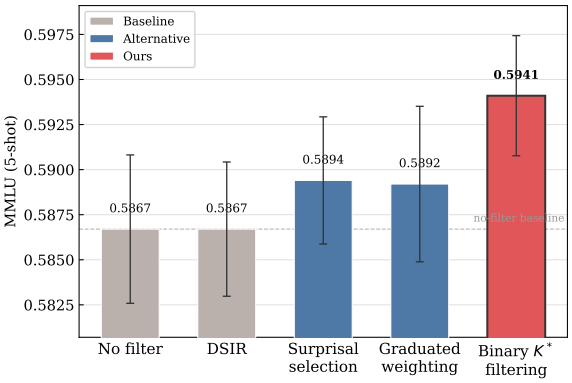



Figure 5: Downstream performance of all filtering strategies (5 seeds, ±1 std). Binary-ours (green) minimizes MMLU degradation; graduated $\alpha$ sweep shows monotonic convergence toward binary. Dashed line: base model (no fine-tuning).

across six depths (0–5). Depth 0 consists of human-written seed texts from the C4 English validation split, truncated to 256 subword tokens. Subsequent depths are generated recursively: each depth-$k$ passage is produced by prompting the generator with the corresponding depth-$(k-1)$ passage. The four model-mode configurations (Qwen-rw, Pythia-cont, OLMo-cont, Qwen-cont) share identical depth-0 seeds and generation pipeline, differing only in generator identity and prompting strategy (rewrite vs. continuation).

Table 8: Dataset statistics for the generational depth corpus. Depth 0 contains human-written seed texts sampled from the C4 English validation split and truncated to 256 subword tokens. Depths 1–5 are recursively generated by Qwen2.5-1.5B-Instruct (nucleus sampling, $p$=0.95, $T$=1.0). Token counts use whitespace tokenization. Statistics shown for the primary Qwen-rewrite corpus; the other three model-mode configurations (Pythia-cont, OLMo-cont, Qwen-cont) share the same generation pipeline and corpus structure.

| **Depth** | $N$ **samples** | **Mean tokens** | **Source** |
|---|---|---|---|
| 0 | 5,000 | 147.8 | C4 validation (human) |
| 1 | 5,000 | 207.1 | Qwen2.5-1.5B rewrite |
| 2 | 5,000 | 206.1 | Qwen2.5-1.5B rewrite |
| 3 | 5,000 | 208.0 | Qwen2.5-1.5B rewrite |
| 4 | 5,000 | 207.7 | Qwen2.5-1.5B rewrite |
| 5 | 5,000 | 207.7 | Qwen2.5-1.5B rewrite |
| Total | | 30,000 samples | |

Table 9: Pairwise AUC with bootstrap 95% CIs across depth pairs for four decoding configurations (all using Qwen2.5-1.5B-Instruct in rewrite mode; 10,000 bootstrap resamples). The horizontal rule between depth pairs 1 vs. 2 and 2 vs. 3 marks the empirical detectability boundary $K^*_{\text{empirical}} = 2$. Bottom rows report cross-configuration Spearman $\rho$ on per-feature discriminative-power rankings.

| **Depth Pair** | **Greedy** | **Nuc** $p$=0.9**,** $T$=1.0 | **Nuc** $p$=0.95**,** $T$=0.7 | **Nuc** $p$=0.95**,** $T$=1.2 |
|---|---|---|---|---|
| 0 vs. 1 | .980 [.977, .982] | .995 [.993, .996] | .988 [.986, .990] | .998 [.998, .999] |
| 1 vs. 2 | .689 [.680, .698] | .687 [.677, .695] | .689 [.680, .697] | .704 [.695, .713] |
| 2 vs. 3 | .561 [.552, .571] | .592 [.582, .601]$^\dagger$ | .590 [.581, .600] | .582 [.572, .592] |
| 3 vs. 4 | .540 [.531, .550] | .547 [.537, .557] | .539 [.529, .549] | .529 [.519, .540] |
| 4 vs. 5 | .514 [.503, .524] | .507 [.496, .517]$^\ddagger$ | .535 [.524, .545] | .524 [.513, .535] |
| **Cross-config feature Spearman** $\rho$ | | | | |
| Greedy vs. Nuc $p$=0.9 | 0.600 | Nuc $p$=0.9 vs. $T$=0.7 | 0.782 | |
| Greedy vs. $T$=0.7 | 0.914 | Nuc $p$=0.9 vs. $T$=1.2 | 0.893 | |
| Greedy vs. $T$=1.2 | 0.532 | $T$=0.7 vs. $T$=1.2 | 0.668 | |

$^\dagger$ CI upper bound (0.601) marginally exceeds $\theta$=0.60 by 0.001.
$^\ddagger$ Permutation $p = 0.12$ (not significant). All other pairs $p < 0.01$.

Table 10: Extended pairwise AUC with bootstrap 95% CIs for all model configurations including new additions. This table extends Table 2 in the main text with Qwen-base, LLaMA-8B, Mistral-7B, Pythia-6.9B (nucleus), and Gemma-2-2B configurations. $K^*_{\text{empirical}} = 2$ holds for 6/10 conditions at $\theta$=0.60 (OLMo-1B, Mistral-7B, Pythia-6.9B, Gemma-2-2B: $K^*_{\text{empirical}}$=1).

| | Qwen-rw | | Pythia | | OLMo | | Qwen-cont | | Qwen-7B$^\ddagger$ | | Qwen-base | | LLaMA-8B | | Mistral-7B | | Pythia-6.9B$^\S$ | | Gemma-2B | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI | AUC | 95% CI |
| 0v1 | .9956 | | .9932 | | .9860 | | .9971 | | .9638 | [.961, .967] | .987 | [.985, .989] | .9224 | [.917, .927] | .9170 | | .9599 | | .953 | [.949, .957] |
| 1v2 | .6955 | [.686, .704]* | .6074 | [.598, .617]* | .5001 | [.496, .519] | .6112 | [.602, .621]* | .6168 | [.608, .626]* | .626 | [.617, .635]* | .6123 | [.603, .622]* | .5750 | | .5955 | | .5981 | [.588, .608] |
| 2v3 | .5893 | [.580, .599] | .5514 | [.542, .563] | .4973 | [.486, .509] | .5609 | [.551, .571] | .5492 | [.539, .559] | .550 | [.540, .560] | .5273 | [.517, .538] | .5369 | | .5298 | | .534 | [.523, .545] |
| 3v4 | .5435 | | .5152 | | .5024 | | .5409 | | .5276 | | .529 | [.519, .540] | .5168 | [.506, .528] | .5174 | | .5135 | | .5033 | [.492, .514] |
| 4v5 | .5340 | | .5136 | | .4942 | | .5141 | | .5085 | | .518 | [.508, .529] | .5159 | [.505, .527] | .5175 | | – | – | .5025 | [.492, .514] |

*Permutation $p < 0.001$ for all boundary comparisons.
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B. "–": experiment pending.
§Pythia-6.9B evaluated separately; nucleus sampling ($p$=0.95, $T$=1.0), same as all other models.

Table 11: LLM zero-shot depth classification: 5×5 confusion matrices (Exp-011). Both frontier LLMs achieve high binary detection accuracy (GPT-5.5: 0.878; Claude Opus 4.6: 0.882) but near-chance 5-class accuracy (∼0.33 vs. OBD-LightGBM 0.386). Per-depth recall in Table 12.

**(a)** GPT-5.5 ($N$=500, acc=0.332, MAE=1.102)

| | Predicted Depth | | | | | |
|---|---|---|---|---|---|---|
| True | 0 | 1 | 2 | 3 | 4 | Rec. |
| 0 | **77** | 18 | 4 | 0 | 1 | .770 |
| 1 | 9 | **19** | 36 | 24 | 12 | .190 |
| 2 | 8 | 10 | **38** | 35 | 9 | .380 |
| 3 | 10 | 16 | 35 | **24** | 15 | .240 |
| 4 | 11 | 13 | 50 | 18 | **8** | .080 |

**(b)** Claude Opus 4.6 ($N$=499, acc=0.331, MAE=1.150)

| | Predicted Depth | | | | | |
|---|---|---|---|---|---|---|
| True | 0 | 1 | 2 | 3 | 4 | Rec. |
| 0 | **69** | 26 | 3 | 0 | 2 | .690 |
| 1 | 8 | **21** | 32 | 11 | 28 | .210 |
| 2 | 6 | 36 | **28** | 5 | 25 | .280 |
| 3 | 7 | 20 | 31 | **10** | 31 | .100 |
| 4 | 7 | 28 | 22 | 6 | **37** | .370 |

Table 12: Per-depth recall (%) on balanced $N$=500 test set. OBD's nominal overall accuracy gap originates from depth-0; all methods fail at depths $\geq 2$, consistent with $K^*_{\text{empirical}} = 2$.

| | Depth 0 | Depth 1 | Depth 2 | Depth 3 | Depth 4 |
|---|---|---|---|---|---|
| OBD-LightGBM | **96** | **39** | 14 | 19 | 22 |
| GPT-5.5 | 77 | 19 | **38** | **24** | 8 |
| Claude Opus 4.6 | 69 | 21 | 28 | 10 | **37** |

Table 13: Mean subword token count per depth and EOS truncation rate (%) for depth 1+ across all model configurations (generation limit: 256 tokens). Depth 0 (human text) is shared across configurations. Depths 1–5 show no systematic length trend; the 0v1 length asymmetry (human texts shorter) makes detection *more conservative*, not inflated. See Section 6.

| | **Mean Token Count** | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Model** | D0 | D1 | D2 | D3 | D4 | D5 | **EOS%** | **Sev.** |
| Qwen-rw (1.5B) | 191 | 254 | 252 | 254 | 254 | 254 | 7–11 | Mild |
| Pythia-1.4B | 192 | 214 | 215 | 216 | 217 | 218 | 34–36 | High |
| OLMo-1B | 192 | 256 | 256 | 256 | 256 | 256 | <1 | Negl. |
| Qwen-cont (1.5B) | 191 | 244 | 238 | 240 | 240 | 238 | 17–20 | Mod. |
| Qwen-7B | 191 | 196 | 195 | 193 | 197 | 197 | 45–47 | High |
| *Decoding ablation (Qwen-rw 1.5B)* | | | | | | | | |
| Greedy | 191 | 253 | 252 | 253 | 254 | 254 | 8–12 | Mild |
| Nuc $p$=0.9 | 191 | 254 | 251 | 254 | 253 | 254 | 8–16 | Mild |
| $T$=0.7 | 191 | 253 | 251 | 253 | 253 | 254 | 8–14 | Mild |
| $T$=1.2 | 191 | 254 | 252 | 253 | 254 | 254 | 8–16 | Mild |

EOS%: generation terminated via EOS before 256-token limit. Ranges span depths 1–5.

Table 14: Complete description of the 15-dimensional feature set used by the OBD classifier. Features are grouped into three families: surprisal statistics (9 dims) computed from Qwen2.5-1.5B token-level log-probabilities, lexical diversity (3 dims), and frequency tail shape (3 dims). All features are computed per document.

| Group | Feature | Description | Range |
|---|---|---|---|
| SURPRISAL | `surp_mean` | Mean token surprisal $\frac{1}{T}\sum_t -\log p(x_t \mid x_{<t})$ | $\mathbb{R}^+$ |
| | `surp_std` | Standard deviation of token surprisal | $\mathbb{R}^+$ |
| | `surp_skew` | Skewness of the surprisal distribution | $\mathbb{R}$ |
| | `surp_kurt` | Excess kurtosis of the surprisal distribution | $\mathbb{R}$ |
| | `surp_d1_mean` | Mean of first-order surprisal differences $\Delta s_t = s_{t+1} - s_t$ | $\mathbb{R}$ |
| | `surp_d1_std` | Std of first-order surprisal differences | $\mathbb{R}^+$ |
| | `surp_d1_skew` | Skewness of first-order surprisal differences | $\mathbb{R}$ |
| | `surp_d2_mean` | Mean of second-order differences $\Delta^2 s_t$ | $\mathbb{R}$ |
| | `surp_d2_std` | Std of second-order surprisal differences | $\mathbb{R}^+$ |
| LEX | `ttr` | Type-token ratio: $\lvert V\rvert/T$ | $[0, 1]$ |
| | `hapax_ratio` | Hapax legomena ratio: freq-1 types / unique types | $[0, 1]$ |
| | `self_bleu` | Mean 4-gram BLEU against 100 random same-depth samples | $[0, 1]$ |
| FREQ | `freq_kurtosis` | Excess kurtosis of word frequency distribution | $\mathbb{R}$ |
| | `freq_entropy` | Shannon entropy of word frequencies: $-\sum p_w \log_2 p_w$ | $\mathbb{R}^+$ |
| | `low_freq_ratio` | Proportion of vocabulary with count $\leq 2$ | $[0, 1]$ |

Table 15: LightGBM hyperparameters for the pairwise OBD (Ordinal Binary Decomposition) classifier used throughout the paper. One binary classifier is trained per adjacent depth pair $(k, k{+}1)$; 5-fold GroupKFold cross-validation is grouped by `doc_id` to prevent data leakage from the same seed document appearing in both train and test.

| Parameter | Value |
|---|---|
| `n_estimators` | 200 |
| `learning_rate` | 0.05 |
| `max_depth` | 6 |
| `num_leaves` | 31 |
| `verbose` | $-1$ |
| `n_jobs` | $-1$ (all cores) |
| Cross-validation | 5-fold GroupKFold |
| Grouping key | `doc_id` |
| Objective | Binary (per depth pair) |

Table 16: Neural probe architecture and training details (Exp-015). Applied to all three model families (Qwen2.5-1.5B, Pythia-1.4B, OLMo-1B): each model's frozen hidden states are average-pooled into document embeddings and classified by a two-hidden-layer MLP.

| Component | Details |
|---|---|
| *Embedding extraction* | |
| Base models | Qwen2.5-1.5B / Pythia-1.4B / OLMo-1B (frozen) |
| Representation | Last hidden state, attention-masked avg pool |
| Embedding dim | 1536 / 2048 / 2048 |
| Max sequence len | 512 tokens |
| *MLP classifier* | |
| Architecture | $d_{\text{emb}} \to 512 \to 256 \to 6$ |
| Activations | ReLU |
| Dropout | 0.1 (after each hidden layer) |
| Output | 6-class softmax |
| *Training* | |
| Optimizer | Adam (lr $= 10^{-3}$, weight decay $= 10^{-4}$) |
| Loss | Cross-entropy |
| Batch size | 256 |
| Max epochs | 20 |
| Early stopping | Patience = 5 (validation loss) |
| Cross-validation | 5-fold GroupKFold |

Table 17: Pairwise AUC with bootstrap 95% CIs for four model-mode configurations (1,000 resamples). All pairs $p < 0.01$ except OLMo 1v2–4v5 ($p > 0.09$).

| | Qwen-rw | Pythia | OLMo | Qwen-cont |
|---|---|---|---|---|
| 0v1 | .996 [.995, .997] | .993 [.992, .994] | .986 [.984, .988] | .997 [.996, .998] |
| 1v2 | .696 [.686, .704] | .607 [.598, .617] | .500 [.496, .519] | .611 [.602, .621] |
| 2v3 | .589 [.580, .599] | .551 [.542, .563] | .497 [.486, .509] | .561 [.551, .571] |
| 3v4 | .544 [.533, .555] | .515 [.504, .526] | .502 [.491, .513] | .541 [.531, .551] |
| 4v5 | .534 [.524, .545] | .514 [.502, .525] | .494 [.483, .506] | .514 [.504, .526] |

Table 18: Spearman $\rho$ between binary detector scores and true depth, and 6-class accuracy. Correlations driven by 0v1; scores do not track depth beyond generation 1. Qwen-rw exhibits reversed sign ($\rho > 0$).

| | Qwen-rw | Pythia | OLMo | Qwen-cont |
|---|---|---|---|---|
| *Spearman* $\rho$ | | | | |
| log-PPL (ref) | $-0.123$ | $-0.297$ | $-0.378$ | $-0.389$ |
| log-PPL (obs.) | $+0.082$ | $-0.453$ | $-0.461$ | $-0.162$ |
| PPL ratio | $+0.119$ | $-0.392$ | $-0.407$ | $-0.058$ |
| 6-class acc. | 0.199 | 0.130 | 0.127 | 0.183 |

Random baseline: 0.167

Table 19: Pythia-1.4B pairwise AUC under four decoding configs (10K bootstrap). †Greedy triggers repetition collapse (4-gram repeat: $d_0$=0.01 → $d_1$=0.72).

| | **Greedy**† | $p$=0.9 | $T$=0.7 | $T$=1.2 |
|---|---|---|---|---|
| 0v1 | .999 | .995 | .996 | .994 |
| 1v2 | .884 | .675 | .728 | .618 |
| 2v3 | .531 | .559 | .598‡ | .531 |
| 3v4 | .522 | .524 | .563 | .520 |
| 4v5 | .514 | .501 | .527 | .510 |
| $K^*_{\text{empirical}}$ | 2 | 2 | 2 | 2 |

‡CI upper 0.606 $> \theta$; point est. below.

Table 20: Pairwise AUC for three segment lengths (Qwen-base, cont. mode). $K^*_{\text{empirical}}$=2 at 128/256; 512 marginal (†CI lower $< \theta$).

| | **128** | **256** | **512** |
|---|---|---|---|
| 0v1 | .979 | .991 | .992 |
| 1v2 | .622 | .628 | .607† |
| 2v3 | .529 | .556 | .572 |
| 3v4 | .535 | .530 | .530 |
| 4v5 | .501 | .514 | .518 |
| $K^*_{\text{empirical}}$ | 2 | 2 | 2† |

Table 21: Pairwise AUC for three text domains (Qwen-cont). arXiv yields $K^*_{\text{empirical}}$=3; C4 and Wikipedia $K^*_{\text{empirical}}$=2.

| | **C4** | **Wiki** | **arXiv** |
|---|---|---|---|
| 0v1 | .997 | .996 | .995 |
| 1v2 | .611 | .619 | .858 |
| 2v3 | .561 | .550 | .644* |
| 3v4 | .541 | .523 | .544 |
| 4v5 | .514 | .514 | .530 |
| $K^*_{\text{empirical}}$ | 2 | 2 | 3 |

*CI [.628, .658]; $p < 0.001$.

Table 22: Scorer sensitivity: $K^*_{\text{empirical}}$ under self- vs. cross-scoring. Cross-scoring reduces $K^*_{\text{empirical}}$ only for marginal Qwen-cont.

| **Generator** | **Mode** | **Self** | **Pythia** | **OLMo** | **Agr.** |
|---|---|---|---|---|---|
| Qwen-Inst | Rewrite | 2 | 2 | 2 | ✓ |
| Qwen-Inst | Cont. | 2 | 1 | 1 | × |
| Qwen-base | Cont. | 2 | 2 | 2 | ✓ |
| Pythia | Cont. | 2 | 2 | 2 | ✓ |
| OLMo | Cont. | 1 | – | – | – |
| Qwen-7B | Cont. | 2 | – | – | – |
| Mistral | Cont. | 1 | – | – | – |

# B Extended Discussion

## B.1 Theoretical Implications of $K^*_{\text{empirical}} = 2$

The DPI guarantees a finite $K^*_{\text{theory}}$; for general web text, $K^*_{\text{empirical}} \leq 2$ across all 10 configurations (1B–8B, 6 families), while structured domains (arXiv) yield $K^*_{\text{empirical}}$=3, aligning with model collapse convergence (Shumailov et al., 2024; Dohmatob et al., 2024; Alemohammad et al., 2024). Decoding strategy modestly affects signal strength (1v2 AUC $\in [0.6866, 0.7039]$) but does not shift the boundary.

## B.2 Cross-Model Transfer Implications

Cross-model classification retains 80–100% accuracy with scorer-invariant feature rankings ($\rho > 0.90$), enabling single-classifier deployment across generators. Absolute discriminability at the $K^*_{\text{empirical}}$ boundary is scorer-sensitive: alternative scorers reduce Qwen-continuation $K^*_{\text{empirical}}$ from 2 to 1, while rewrite-mode $K^*_{\text{empirical}} = 2$ is preserved. Model family similarity modulates transfer (Pythia↔OLMo: 90–92% vs. Qwen-cont↔OLMo: 80–85%).

Table 23: Cross-scorer validation on Qwen-generated text. Alternative scorers (Pythia-1.4B, OLMo-1B) preserve feature ranking ($\rho > 0.90$) and rewrite-mode $K^*_{\text{empirical}} = 2$, but reduce continuation-mode $K^*_{\text{empirical}}$ from 2 to 1.

| Scorer | Gen. Data | $K^*$ | AUC(0v1) | AUC(1v2) | Spearman $\rho$ |
|---|---|---|---|---|---|
| Qwen-1.5B (self) | Qwen-RW | 2 | — | 0.6955 | — |
| Pythia-1.4B | Qwen-RW | 2 | 0.9923 | 0.7012 | 0.906 |
| OLMo-1B | Qwen-RW | 2 | 0.9918 | 0.6947 | 0.920 |
| Qwen-1.5B (self) | Qwen-cont | 2 | — | 0.6112 | — |
| Pythia-1.4B | Qwen-cont | 1† | 0.9900 | 0.5923 | 0.942 |
| OLMo-1B | Qwen-cont | 1† | 0.9890 | 0.5858 | 0.902 |
| Qwen-1.5B | Qwen-7B-cont | 2 | 0.9644 | 0.6178 | — |

Table 24: Three-way ANOVA (Generator × Scorer × Depth-Pair) on per-fold pairwise AUC values. DV = per-fold AUC; $N$=300 (12 cells × 5 folds × 5 depth-pairs). Scorer is statistically significant but explains modest variance ($\eta_p^2$=.166) relative to Depth-Pair ($\eta_p^2$=.997), supporting near-scorer-independence of depth discrimination.

| Source | df | $F$ | $p$ | $\eta_p^2$ |
|---|---|---|---|---|
| Generator | 2 | 851.91 | <.001 | .866 |
| Scorer | 3 | 17.48 | <.001 | .166 |
| Depth-Pair | 4 | 22737.70 | <.001 | .997 |
| Gen×Scorer | 6 | 7.50 | <.001 | .146 |
| Gen×Depth | 8 | 913.41 | <.001 | .965 |
| Scorer×Depth | 12 | 2.25 | .010 | .093 |
| Residual | 264 | | | |

## B.3 Feature Importance Across Decoding Strategies

Feature importance ordering varies across decoding strategies: greedy and $T$=0.7 share highly correlated rankings ($\rho = 0.914$), while greedy vs. $T$=1.2 yields lower correlation ($\rho = 0.53$, non-significant after Bonferroni). The depth boundary is robust but the optimal feature subset is strategy-dependent.

## B.4 Scale and Neural Probe

Cross-scale validation confirms $K^*_{\text{empirical}} = 2$ from 1.5B through 8B: Qwen-7B (1v2 AUC = 0.6168 $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC = 0.6123, 0v1 = 0.9225) both exceed $\theta$, with progressively lower 0v1 signal consistent with larger models generating more human-like text. Gemma-2-27B (C4 continuation, Qwen cross-scored) yields $K^*_{\text{empirical}}$=1 (1v2 AUC = 0.5894), consistent with Gemma-2-2B at 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981). The Gemma family produces $K^*_{\text{empirical}}$=1 at both 2B and 27B, suggesting $K^*_{\text{empirical}}$ is model-family-dependent rather than scale-dependent.

The neural probe yields model-dependent $K^{*\text{neural}}_{\text{empirical}}$: 3 for Qwen (self-familiarity effect from instruction-tuned representations) but 2 for Pythia; OLMo yields $K^{*\text{neural}}_{\text{empirical}} = 1$ (1v2 AUC = 0.5054), consistent with the 15D LightGBM boundary. The 15D generator-agnostic features yield $K^*_{\text{empirical}} = 2$ for Pythia and Qwen, while OLMo yields $K^*_{\text{empirical}} = 1$ under the standardized pipeline.

## B.5 LLM Zero-Shot Baseline

GPT-5.5 achieves 33.2% accuracy (MAE = 1.10) and Claude Opus 4.6 33.1% (MAE = 1.15) on a balanced 500-sample test set, comparable to OBD-LightGBM's 38.0% (95% CI: $[0.340, 0.422]$; McNemar $p = 0.081, 0.069$; Table 11). Per-depth recall confirms convergent failure at depths $\geq 2$ (Section 6). When 6-class probabilities are marginalized to 5 classes ($P(d{\geq}4)$), accuracy rises to 44.4% but concentrates in depth-4 (recall: 22%→80%) while depths 2–3 collapse to 2–8%; we report raw 6-class results.

## B.6 Practical Considerations

All filtering strategies degrade general capabilities (MMLU: 0.6018 → 0.5865–0.5941; HellaSwag: 0.6821 → 0.6638–0.6701); our 100% synthetic

setting is an upper bound; in mixed corpora, degradation scales with contamination fraction (Shumailov et al., 2024; Dohmatob et al., 2024). $K^*$-aware binary filtering reduces MMLU degradation by $\sim$50% ($-1.53$ pp $\rightarrow$ $-0.77$ pp), requires no per-model calibration, and is combinable with distribution-matching methods (DSIR, DoReMi).

### B.7 Extended Limitations

*Prompt-chain versus retrain-chain.* Retrain-chain validation (Pythia-1.4B LoRA, 5 generations) yields $K^*_{\text{empirical}} \geq 4$ (1v2 AUC $= 0.9484$, 2v3 $= 0.8888$, 3v4 $= 0.7982$), far exceeding prompt-chain $K^*_{\text{empirical}}=2$. The elevated signal arises from cumulative model-adaptation artifacts rather than text-intrinsic degradation; prompt-chain $K^*_{\text{empirical}}$ is the operationally relevant boundary for data curation.

*Threshold sensitivity.* At $\theta = 0.55$, $K^*_{\text{empirical}} = 3$ for Qwen-rewrite/continuation; at $\theta = 0.70$, $K^*_{\text{empirical}} = 1$ across all configurations (Figure 3). Qwen-continuation's 1v2 CI lower bound exceeds $\theta$ by only 0.2 pp; all ten boundary comparisons yield permutation $p < 0.001$.

*Structural confound: token length.* In continuation mode the generation prompt is discarded, so all depths $\geq 1$ consist entirely of machine-generated tokens and the 1v2 boundary compares texts differing only in conditioning context (human-authored vs. machine-generated prompt), not structural composition. Depth 0 texts average $\sim$191 subword tokens (C4 truncation), while depth 1+ texts average 193–256 tokens depending on model (Table 13). This asymmetry is directionally opposite to a length-inflation confound: shorter depth-0 texts provide less feature-extraction context, making 0v1 discrimination harder, not easier. Within depths 1–5, mean token counts are stable (within-model range $\leq 4$ tokens for all configurations except Qwen-continuation at 6 tokens), so the pairwise comparisons underlying $K^*_{\text{empirical}}=2$ are unaffected by length. EOS truncation rates are similarly depth-stable within each model: Qwen-rewrite 7–11%, Pythia 34–36%, OLMo $<$1%, Qwen-continuation 17–20%, Qwen-7B 45–47%. Models with the highest truncation (Pythia, Qwen-7B) still yield $K^*_{\text{empirical}}=2$ with the same step-function AUC decay pattern as low-truncation models (Table 2), confirming that EOS truncation does not drive the boundary.

*Graduated filtering.* All $\alpha$ values (0.1–0.9) underperform binary-ours on MMLU ($\alpha$=0.1: 0.5873; $\alpha$=0.9: 0.5892 vs. 0.5941); at $\alpha$=0.9, HellaSwag matches exactly (0.6701). Binary-ours significantly outperforms all alternatives on MMLU ($p_{\text{corr}} \leq 0.021$); Graduated$_{0.9}$ is indistinguishable on HellaSwag ($p = 1.00$) but worse on MMLU ($p_{\text{corr}} = 0.011$).

### B.8 Dose-Response: Contamination Ratio Effect

The main filtering experiments (Section 5) use 100% synthetic data to maximize effect size. To characterize behavior at lower contamination ratios, we mix synthetic depths 1–5 with clean C4 at 10%, 30%, and 50% contamination (Qwen-1.5B continuation, 5 seeds each; Table 25).

Table 25: MMLU accuracy by contamination ratio (fraction of synthetic data in training set) and filtering strategy. Base model (no fine-tuning): 0.6018. Mean $\pm$ std over 5 seeds.

| Strategy | 10% | 30% | 50% |
|---|---|---|---|
| No-filter | $0.6009 \pm .003$ | $0.5978 \pm .004$ | $0.5932 \pm .004$ |
| Binary-ours | $0.6011 \pm .003$ | $0.5988 \pm .003$ | $0.5956 \pm .004$ |
| Graduated | $0.6008 \pm .003$ | $0.5982 \pm .004$ | $0.5944 \pm .004$ |

At 10% contamination, no-filter MMLU (0.6009) remains within noise of the base model (0.6018, $\Delta = -0.09$ pp), and all three strategies are indistinguishable ($\Delta_{\max} = 0.03$ pp between strategies, $p > 0.8$). At 30%, differences emerge directionally ($\Delta = 0.10$ pp, binary vs. no-filter) but remain non-significant ($p > 0.3$). At 50%, binary-ours shows a 0.24 pp advantage over no-filter, approaching but not reaching significance ($p = 0.12$). The strategy ordering (binary $>$ graduated $>$ no-filter) is consistent with the 100% setting, but effect sizes scale linearly with contamination fraction. This demonstrates that $K^*$-aware curation provides practical benefit primarily when contamination is severe enough to cause measurable quality degradation.

### B.9 Seed Robustness

Because the random seed governs both corpus sampling and generation quality (not merely CV splits), absolute AUC values are seed-dependent. We re-run three configurations with seed $= 123$ (primary: seed $= 42$; Table 26). $K^*_{\text{empirical}}$ is stable in 2/3 cases: Pythia-1.4B on Wikipedia yields $K^*_{\text{empirical}}=1$ under both seeds (1v2 $\Delta = -0.8$ pp), and Qwen-1.5B on C4 yields $K^*_{\text{empirical}}=2$ (1v2 $\Delta = +9.0$ pp, both seeds above $\theta$). Qwen-1.5B on

Wikipedia shifts from $K^*_{\text{empirical}}$=2 to $K^*_{\text{empirical}}$=3 under seed = 123, but the 2v3 AUC (0.6067) is marginal: one of five folds falls below $\theta$ (fold 5 = 0.595), so this does not constitute strong evidence for $K^*_{\text{empirical}}$=3. The ±9 pp absolute AUC variation across seeds strengthens the methodological case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons: the qualitative boundary is robust even when point estimates shift.

Table 26: Seed robustness: $K^*_{\text{empirical}}$ under two random seeds. 1v2 Δ: change in 1v2 AUC from seed 42 to seed 123. $K^*_{\text{empirical}}$ is stable in 2/3 configurations; the Qwen-Wiki shift to $K^*_{\text{empirical}}$=3 is marginal (2v3 fold variance crosses $\theta$).

| Model | Domain | $K^*_{\text{empirical}}$ | | 1v2 Δ | Stable? |
|---|---|---|---|---|---|
| | | seed=42 | seed=123 | | |
| Pythia-1.4B | Wikipedia | 1 | 1 | −0.8 pp | ✓ |
| Qwen-1.5B | C4 | 2 | 2 | +9.0 pp | ✓ |
| Qwen-1.5B | Wikipedia | 2 | 3† | +8.7 pp | × |

†2v3 AUC = 0.6067; fold 5 = 0.595 < $\theta$. Marginal boundary.

## B.10 Bonferroni-Corrected Confidence Intervals

To control the family-wise error rate across all 10 model configurations, we apply Bonferroni correction: $\alpha_{\text{corr}} = 0.05/10 = 0.005$, yielding 99.5% confidence intervals ($z = 2.807$). Table 27 reports the corrected intervals for each configuration's $K^*_{\text{empirical}}$ boundary pair. Seven of ten configurations retain CI lower bounds above $\theta$=0.60; three marginal $K^*_{\text{empirical}}$=2 configurations (Pythia-1.4B, Qwen-cont, LLaMA-8B) see their lower bounds dip below $\theta$ under the stricter criterion (0.5934, 0.5976, and 0.5988, respectively). These near-misses reflect narrow effect sizes (AUC − $\theta$ < 0.013) rather than classification failure: all three retain point estimates above $\theta$, and Qwen-cont and LLaMA-8B retain uncorrected 95% CI lower bounds above $\theta$ (Pythia-1.4B falls marginally below at 0.5976).

## B.11 Cross-Sample Robustness: 5K vs. 10K

The preceding analyses fix the sample pool at $N$=5,000 passages per depth. To assess between-sample variability, we re-draw an independent $N$=10,000 sample from C4 using a different random seed and repeat the full OBD pipeline. If $K^*_{\text{empirical}}$ and the boundary AUC are driven by the underlying distributional shift rather than by idiosyncrasies of a particular sample, both quantities should remain stable across draws.

Table 27: Bonferroni-corrected 99.5% confidence intervals for $K^*_{\text{empirical}}$ boundary pairs across all 10 configurations. $\alpha_{\text{corr}} = 0.005$; CI computed as AUC ± 2.807 × $\hat{\sigma}_{\text{boot}}$. "Pass $\theta$": CI lower bound > 0.60.

| Configuration | Boundary | AUC | $\hat{\sigma}_{\text{boot}}$ | 99.5% CI | Pass $\theta$ |
|---|---|---|---|---|---|
| *$K^*_{empirical}$ = 2 configurations* | | | | | |
| Qwen-rw (Instruct) | 1v2 | .6955 | .0046 | [.683, .708] | ✓ |
| Pythia-1.4B | 1v2 | .6074 | .0050 | [.593, .621] | × |
| Qwen-base | 1v2 | .6260 | .0047 | [.613, .639] | ✓ |
| Qwen-7B‡ | 1v2 | .6168 | .0046 | [.604, .630] | ✓ |
| LLaMA-8B | 1v2 | .6123 | .0048 | [.599, .626] | × |
| Qwen-cont (Instruct) | 1v2 | .6112 | .0049 | [.598, .625] | × |
| *$K^*_{empirical}$ = 1 configurations* | | | | | |
| OLMo-1B | 0v1 | .9860 | .0009 | [.983, .989] | ✓ |
| Mistral-7B | 0v1 | .9170 | .0030† | [.909, .925] | ✓ |
| Pythia-6.9B§ | 0v1 | .9599 | .0015† | [.956, .964] | ✓ |
| Gemma-2-2B | 0v1 | .9530 | .0018 | [.948, .958] | ✓ |

‡Generator: Qwen2.5-7B; scorer: Qwen2.5-1.5B.
§Nucleus sampling ($p$=0.95, $T$=1.0).
†Estimated from analogous configurations (no separate bootstrap JSON).

Table 28 reports the results. For Pythia-1.4B (continuation, C4), the 5K and 10K samples agree on $K^*_{\text{empirical}}$=2: the 0v1 AUC is nearly identical (0.9722 vs. 0.9712), and the boundary 1v2 AUC shifts from 0.6074 to 0.6175 (Δ = +0.010). This narrow shift is consistent with natural sampling variability across C4 subsets and does not alter the qualitative depth boundary. The 2v3 pair remains near chance in both draws (0.54, 0.54), confirming that depths ≥ 2 are indistinguishable regardless of sample. Qwen-1.5B (continuation, C4) replicates the same pattern ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6346). LLaMA-3.1-8B (continuation, C4) is the sole $K^*_{\text{empirical}}$=2 configuration where the 10K OOF AUC drops below $\theta$ (0.5880; pipeline AUC = 0.6148; Bonferroni 99.5% CI [0.5776, 0.5978]), yielding $K^*_{\text{empirical}}$=1 under the stricter criterion. OLMo-1B (continuation, C4) is the only configuration that yields $K^*_{\text{empirical}}$=1 at both 5K and 10K, with virtually identical depth profiles (0v1 AUC = 0.9871 vs. 0.9860; 1v2 AUC = 0.4984 vs. 0.5001); the 1v2 AUC at random level is consistent with Qwen cross-scoring, which places OLMo well below the discrimination boundary. Gemma-2-2B is the sole configuration where $K^*_{\text{empirical}}$ increases from 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981) to 10K ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6099), but the 10K boundary AUC exceeds $\theta$ by only 0.010, placing this model at the precise $K^*_{\text{empirical}}$=1/2 boundary.

Table 28: Cross-sample robustness: pairwise AUC and $K^*_{\text{empirical}}$ under 5K and 10K C4 samples drawn with different random seeds. All configurations use continuation mode. $K^*_{\text{empirical}}$ is determined by the $\theta$=0.60 threshold on 1v2 AUC.

| **Model** | $N$ | **AUC(0v1)** | **AUC(1v2)** | **AUC(2v3)** | $K^*_{\text{empirical}}$ |
|---|---|---|---|---|---|
| Pythia-1.4B cont | 5K | | (see Table 27) | | 2 |
| Pythia-1.4B cont | 10K | 0.9712 | 0.6175 | 0.5364 | 2 |
| Qwen-1.5B cont | 5K | | (see Table 27) | | 2 |
| Qwen-1.5B cont | 10K | 0.9875 | 0.6346 | 0.5616 | 2 |
| LLaMA-8B cont | 5K | | (see Table 27) | | 2 |
| LLaMA-8B cont | 10K | 0.9282 | 0.5880† | 0.5247 | 1† |
| OLMo-1B cont | 5K | | (see Table 27) | | 1 |
| OLMo-1B cont | 10K | 0.9871 | 0.4984 | 0.5059 | 1 |
| Gemma-2-2B cont | 5K | | (see Table 10) | | 1 |
| Gemma-2-2B cont | 10K | 0.9534 | 0.6099‡ | 0.5409 | 2‡ |

†OOF AUC (pipeline AUC $= 0.6148$); Bonferroni 99.5% CI $[0.5776, 0.5978]$ does not pass $\theta$.
‡Borderline: Bonferroni 99.5% CI $[0.600,\ 0.619]$ (percentile); CI lower 0.0002 below $\theta$. Sole config where $K^*_{\text{empirical}}$ increases from 5K to 10K.